

0511

18. $H_0: p_1 = p_2 = p_3$

$H_a: p_1, p_2, p_3$ 不完全相等.

$$\bar{z}_i = \frac{200}{3} = 66.67$$

$$\chi^2 = \sum_{i=1}^3 \frac{(O_i - \bar{z}_i)^2}{\bar{z}_i} = \frac{(83 - 66.67)^2}{66.67} + \frac{(61 - 66.67)^2}{66.67} + \frac{(58 - 66.67)^2}{66.67} \approx 6.19$$

$$\chi^2_{1,0.05} = 5.9915 < 6.19 \Rightarrow \text{Reject } H_0$$

$\Rightarrow$ 三個入口的偏好有顯著差異.

$$95\% \text{ CI} = \hat{p} \pm z \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$= \frac{83}{200} \pm 1.96 \sqrt{\frac{0.415(1-0.415)}{200}} = [0.346, 0.4833]$$

22. $H_0: p_1 = p_2 = p_3 = \dots = p_7$

$H_a: p_1, p_2, p_3, \dots, p_7$ 不完全相等.

$$\bar{z}_i = \frac{200}{7} = 28.57$$

$$\chi^2 = \sum_{i=1}^7 \frac{(O_i - \bar{z}_i)^2}{\bar{z}_i} = \sum_{i=1}^7 \frac{O_i^2}{\bar{z}_i} - n$$

$$= \frac{24^2 + 36^2 + 27^2 + 26^2 + 32^2 + 26^2 + 29^2}{28.57} - 200$$

$$\approx 3.64$$

$$\chi^2_{6,0.05} = 12.59 > 3.64 \Rightarrow \text{non-Reject } H_0$$

$\Rightarrow$ 沒有足夠證據支持心臟病發作機率會受星期影響.

18. H_0 : 致命事故發生與車輛大小無關 (獨立)
 H_a : 致命事故發生與車輛大小有關 (不獨立)

	S	M	L	Total
F	67	26	16	109
NT	128	63	46	237
Total	195	89	62	346

$$\hat{E}_{F.S} = \frac{109 \times 195}{346}$$

$$\hat{E}_{F.M} = \frac{109 \times 89}{346}$$

$$\hat{E}_{F.L} = \frac{109 \times 62}{346}$$

$$\hat{E}_{NT.S} = \frac{237 \times 195}{346}$$

$$\hat{E}_{NT.M} = \frac{237 \times 89}{346}$$

$$\hat{E}_{NT.L} = \frac{237 \times 62}{346}$$

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - \hat{E}_{ij})^2}{\hat{E}_{ij}} = \sum_{i=1}^r \sum_{j=1}^c \frac{O_{ij}^2}{\hat{E}_{ij}} - n$$

$$= \frac{67^2}{61.43} + \frac{26^2}{28.04} + \frac{16^2}{19.53} + \frac{128^2}{133.57} + \frac{63^2}{60.96} + \frac{46^2}{42.47} - 346 = 1.885$$

$\chi^2_{(31)(2-1), 0.05} = 5.99 > 1.885 \Rightarrow \text{Non-Reject } H_0$
 $\Rightarrow$ 沒有足夠證據顯示致命事故與車輛大小有關

0511.

23 a. H₀: 薪水與在家工作天數無關 (獨立)H_a: 薪水與在家工作天數有關 (不獨立)

Salary	< 1	1 ≤, < All	All	Total
A	38	16	14	68
B	54	26	12	92
C	35	22	9	66
D	33	29	12	74

$$\begin{aligned}
 E_{A.1} &= \frac{68 \times 160}{300} & E_{A.2} &= \frac{68 \times 93}{300} & E_{A.3} &= \frac{68 \times 47}{300} \\
 E_{B.1} &= \frac{92 \times 160}{300} & E_{B.2} &= \frac{92 \times 93}{300} & E_{B.3} &= \frac{92 \times 47}{300} \\
 E_{C.1} &= \frac{66 \times 160}{300} & E_{C.2} &= \frac{66 \times 93}{300} & E_{C.3} &= \frac{66 \times 47}{300} \\
 E_{D.1} &= \frac{74 \times 160}{300} & E_{D.2} &= \frac{74 \times 93}{300} & E_{D.3} &= \frac{74 \times 47}{300}
 \end{aligned}$$

$$\begin{aligned}
 \chi^2 &= \sum_{i=1}^4 \sum_{j=1}^3 \frac{O_{ij}^2}{E_{ij}} - n = \frac{38^2}{36.27} + \frac{16^2}{21.08} + \frac{14^2}{10.65} + \frac{54^2}{49.07} + \frac{26^2}{28.52} + \frac{12^2}{14.41} \\
 &\quad + \frac{35^2}{35.20} + \frac{22^2}{20.146} + \frac{9^2}{10.34} + \frac{33^2}{39.47} + \frac{29^2}{22.94} + \frac{12^2}{11.59} - 300 \approx 6.446
 \end{aligned}$$

$$\chi^2_{(4-1)(3-1), 0.05} = 12.59 > 6.446 \Rightarrow \text{N/A} - \text{Reject } H_0$$

⇒ 沒有足夠證據顯示薪水與在家工作天數有關

$$b. p\text{-value} = P(\chi^2 > 6.45) > P(\chi^2 > 10.645) = 0.1$$

$$0.1 > 0.05 \Rightarrow \text{N/A} - \text{reject } H_0, \text{ 同 (a)}$$